

A differentiable, bit-exact game console as a ground-truth testbed for interpretability

The subject IS the substrate — a fully known program — so every explanation can be scored against the true causes.

1 SUBSTRATE

Differentiable VCS

jutari (Julia) · jaxtari (JAX)

bit-exact to xitari · 64/64 games

- ✓ **Exact forward**
deterministic re-run
- ✓ **Exact interventions**
do(u) on RAM / register / ROM / input
- ✓ **Full observability**
every cell, register, opcode, pixel
- ✓ **Content-path gradients**
 $\partial y / \partial u$ via SOFT-STE

2 ORACLE (§3.2)

Exact intervention oracle

PRIMARY ground truth

do(u:=v') → Exact re-run → Δy
occlude bit-exact $= y' - y_0$

true causal map { $\Delta y(u)$ } · no world model assumed

Content-path gradient companion

$\partial y / \partial u$ via SOFT-STE · raw grad · IG

secondary check — cross-validates the oracle

valid for CONTENT (colour / graphics-bit) outputs

Caveat — discrete-index outputs

sprite position / index / event

naive gradient = 0 ✗

→ the intervention oracle is the SOLE ground truth there; gradient methods are scored AS methods under test.

3 SCORE

any XAI method → explanation $\hat{E}$

method under test

F: causes

claimed causes = TRUE causes

S: behavior

predicts y under held-out do(u)

M: parsimony

minimal, right level of state

right = $F \wedge S \wedge M$

verified against the true causes

axes: faithfulness vs plausibility proxy

danger zone: convincing but not faithful